

Wave Optics, solved

Step-by-step solutions to every practice problem in Chapter 23. Try each problem on your own before you read its solution.

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P23.1

Interference fringes appear only where two waves keep a **fixed phase relationship**, which is called coherence. Light from an ordinary bulb is emitted by countless atoms in short, independent bursts, so its phase jumps randomly every few nanoseconds. Light from two separate bulbs therefore has a phase difference that changes randomly about a billion times a second. The fringes, if any, shift faster than any eye or camera can follow, and they average to uniform illumination.

When two slits are lit by the *same* laser beam, both carry copies of the same wave. Whatever the source's phase does, it does to both, so their phase *difference* at any point on the screen stays fixed. The fringes then stand still and are clearly visible.

P23.2

The colours come from interference between light reflected from the film's front and back surfaces. The path difference, $2nt$, depends on the thickness, and so does which colour interferes constructively. Gravity drains the film, so it is thinnest at the top.

When the top becomes much thinner than a wavelength ($t \ll \lambda$), the path difference is essentially zero for every colour. Only the π phase change remains: there is a phase change at the front surface (air to soap) but not at the back (soap to air). The two reflections therefore cancel for *all* wavelengths, and that region reflects no light and looks **black**. A black film is only a few tens of nanometres thick, so it is about to burst.

P23.3

Diffraction is significant when the size of an obstacle or opening is comparable to the wavelength, or smaller. Radio waves have wavelengths from about a metre to kilometres, comparable to buildings and hills, so they diffract strongly and bend into the “shadow” region behind them. Light has a wavelength of about 5×10^{-7} m, millions of times smaller than everyday objects, so diffraction at their edges is tiny, and the shadows look sharp. Look very closely at the edge of a shadow cast by a point source, however, and you will see faint diffraction fringes.

P23.4

Light reflected from horizontal surfaces such as roads and water is strongly polarised **horizontally**, completely so at Brewster’s angle. Polarised sunglasses have their transmission axis **vertical**, so they block most of this glare while letting through half of the unpolarised light from everything else.

If you tilt your head by 90° , the transmission axis becomes horizontal. The glare passes straight through and the road looks much brighter. You can use this as a simple test of whether sunglasses are polarised.

P23.5

(a) Wavelength:

$$\lambda = \frac{d \Delta y}{L} = \frac{(4.0 \times 10^{-4})(2.5 \times 10^{-3})}{1.5} = 6.7 \times 10^{-7} \text{ m} = 667 \text{ nm (red)}.$$

(b) In water, the wavelength shrinks to $\lambda/n = 667/1.33 = 501$ nm, so the spacing shrinks in proportion:

$$\Delta y' = \frac{2.5}{1.33} = 1.9 \text{ mm}.$$

P23.6

Phase changes. Air to oil ($1.00 \rightarrow 1.45$): reflection from a higher index, so a phase change of π . Oil to water ($1.45 \rightarrow 1.33$): reflection from a lower index, so no phase change. Exactly one reflection has a phase change.

Constructive reflection:

$$2nt = \left(m + \frac{1}{2}\right)\lambda \Rightarrow \lambda = \frac{2(1.45)(300)}{m + \frac{1}{2}} = \frac{870 \text{ nm}}{m + \frac{1}{2}}.$$

m	λ
0	1740 nm (infrared)
1	580 nm (yellow)
2	348 nm (ultraviolet)

The only strongly reflected visible wavelength is **580 nm**, so the film looks **yellow**. As a check, destructive reflection ($2nt = m\lambda$) occurs at 435 nm, so violet-blue is suppressed, which reinforces the yellow appearance.

P23.7

(a) **Minimum thickness.** Both reflections have a phase change (air to MgF_2 , and MgF_2 to glass, both towards higher index). Destructive interference needs $2nt = \frac{1}{2}\lambda$:

$$t = \frac{\lambda}{4n} = \frac{550}{4(1.38)} = 99.6 \text{ nm}.$$

(b) Next thickness: $2nt = \frac{3}{2}\lambda$, so $t = 3\lambda/4n = 299 \text{ nm}$.

(c) **Purplish tint.** The coating cancels reflection best near 550 nm, in the green, at the centre of the visible spectrum. Red and blue light, farther from 550 nm, are cancelled less completely, so the faint remaining reflection is a mixture of red and blue: magenta or purple.

P23.8

$$a = \frac{\lambda}{\sin \theta} = \frac{589 \times 10^{-9}}{\sin 1.50^\circ} = \frac{589 \times 10^{-9}}{0.0262} = 2.25 \times 10^{-5} \text{ m} = 22.5 \mu\text{m}.$$

P23.9

(a) Angular resolution:

$$\theta_{\min} = 1.22 \frac{5.0 \times 10^{-7}}{2.4} = 2.5 \times 10^{-7} \text{ rad} \approx 0.05 \text{ arcsec.}$$

(b) On the Moon:

$$s = L\theta_{\min} = (3.84 \times 10^8)(2.54 \times 10^{-7}) = 98 \text{ m.}$$

The smallest resolvable feature is about 100 m across, so a 10 m feature is **not** resolved. This is why Hubble cannot photograph the Apollo landing sites. Lunar orbiters, a few tens of kilometres above the surface, can.

P23.10

$$d = 1/500 \text{ mm} = 2.0 \text{ } \mu\text{m}, \text{ so } \sin \theta = m\lambda/d.$$

(a) Angular ranges:

- First order: **sin θ** from **0.20** to **0.35**, so **θ** from **11.5°** to **20.5°**.
- Second order: **sin θ** from **0.40** to **0.70**, so **θ** from **23.6°** to **44.4°**.

(b) **Overlap.** The third order starts at **sin $\theta = 3(0.400)/2.0 = 0.60$** , that is **36.9°**. The red end of the third order would need **sin $\theta = 1.05$** , which is impossible, so it is missing. The second order extends to **44.4°**, so the two orders **overlap** between **36.9°** and **44.4°**. In general, second-order red (700 nm) overlaps third-order violet (467 nm), since **2 × 700 = 3 × 467**. Spectrometers use filters to block unwanted orders.

P23.11**Three polarisers:**

1. The first halves the unpolarised light: $I_0/2$.
2. The middle one, at **45°**: $\frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4}$.
3. The last one, at a further **45°**: $\frac{I_0}{4} \cos^2 45^\circ = \frac{I_0}{8}$.

Why adding a filter helps. Crossed polarisers block everything, because the light leaving the first has *no* component along the axis of the second. The middle polariser does not just absorb light. It *re-polarises* the light along its own axis, at 45° to the last one. That new polarisation does have a component along the final axis. Polarisers project the field onto their axis. They are not simple absorbers, so their order and orientation matter.

P23.12

The rotation is $\alpha = [\alpha] \ell c$, so

$$c = \frac{\alpha}{[\alpha] \ell} = \frac{1.05^\circ}{(52.7^\circ \text{ dm}^{-1} (\text{g/mL})^{-1})(1.0 \text{ dm})} = 0.020 \text{ g/mL} = 2.0 \text{ g per 100 mL}.$$

This is **clinically significant**. Glucose appears in the urine when the blood glucose exceeds the kidney's threshold (about 10 mmol/L), so this level suggests poorly controlled diabetes. Clinical labs now usually measure glucose by enzymatic test strips, but polarimetry remains important in the pharmaceutical and sugar industries.

P23.13

(a) Two waves. Add $E_1 = E_0 \sin \omega t$ and $E_2 = E_0 \sin(\omega t + \phi)$:

$$E = 2E_0 \cos \frac{\phi}{2} \sin \left(\omega t + \frac{\phi}{2} \right).$$

The amplitude is $2E_0 \cos(\phi/2)$, and intensity is proportional to amplitude squared:

$$I = 4I_1 \cos^2 \frac{\phi}{2}, \quad \phi = \frac{2\pi d \sin \theta}{\lambda} \Rightarrow I = I_{\max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right).$$

(b) **Missing orders.** Interference maxima are at $d \sin \theta = m\lambda$. Single-slit zeros are at $a \sin \theta = m'\lambda$. When $d = 4a$, the interference maximum $m = 4m'$ falls exactly on a diffraction zero, so it has zero intensity. The 4th, 8th, ... fringes are missing.

Inside the central diffraction maximum ($|a \sin \theta| < \lambda$, so $|m| < 4$), the bright fringes are $m = 0, \pm 1, \pm 2, \pm 3$: 7 fringes.

P23.14

(a) **Dark contact line.** Where the plates touch, the air gap has zero thickness, so there is no extra path. Light reflected from the bottom of the top plate (glass to air) has no phase change. Light reflected from the top of the bottom plate (air to glass) has a π change. They cancel, so the contact line is dark.

(b) **Hair thickness.** For an air gap of thickness t , the dark fringes are at $2t = m\lambda$. Twenty dark fringes, counting the contact line as $m = 0$, means the hair is at about $m = 19$:

$$D = \frac{19\lambda}{2} = \frac{19(600 \text{ nm})}{2} = 5.7 \text{ }\mu\text{m}.$$

(If the hair sits between dark fringes, the uncertainty is about $\pm\lambda/4$, which is $\pm 0.15 \text{ }\mu\text{m}$.) That is thinner than a human hair, which is typically 50–100 μm , so this would be a very fine fibre.

(c) **Fringe spacing:**

$$\Delta x = \frac{\lambda L}{2D} = \frac{(600 \times 10^{-9})(0.10)}{2(5.7 \times 10^{-6})} = 5.3 \text{ mm}.$$

This equals $L/19$, as expected.

P23.15

(a) **Coherence length:**

$$l_c \approx 0.44 \frac{\lambda_0^2}{\Delta\lambda} = 0.44 \frac{(840 \text{ nm})^2}{50 \text{ nm}} = 6.2 \times 10^3 \text{ nm} = 6.2 \text{ }\mu\text{m}.$$

(b) **In tissue,** path lengths are scaled by n , so the resolution is $6.2/1.38 = 4.5 \text{ }\mu\text{m}$. That is enough to distinguish the roughly ten cellular layers of the retina.

(c) **Bandwidth.** A broad spectrum is made of many wavelengths. They stay in step only over a short distance before they drift out of phase. A broader bandwidth means faster dephasing, and so a shorter coherence length (the time–frequency, or Fourier, trade-off). Interference then occurs only for path lengths matched very closely, which pins down the depth precisely. Ultra-high-resolution OCT uses sources with bandwidths above 100 nm to reach 2–3 μm .

(d) **Fringes.** Moving the mirror by 0.25 μm changes the path by twice that, 0.50 μm , which is exactly one wavelength. So **one** fringe passes.

P23.16

(a) Phone at 30 cm:

$$s = (0.30 \text{ m})(2.9 \times 10^{-4}) = 87 \mu\text{m}, \quad \frac{25.4 \text{ mm}}{0.087 \text{ mm}} \approx 290 \text{ pixels per inch.}$$

(b) Television at 3.0 m:

$$s = (3.0)(2.9 \times 10^{-4}) = 0.87 \text{ mm} \Rightarrow \approx 29 \text{ pixels per inch.}$$

Comment. About 300 ppi at phone distance is the basis of “retina display” claims. Beyond that density, individual pixels cannot be distinguished by a typical eye, although people with better than average acuity can see slightly finer detail. A 55-inch 4K television has about 80 ppi, which is more than enough at 3 m. Higher resolution only helps if you sit much closer or have exceptional vision.

P23.17(a) Bragg angles, from $2d \sin \theta = m\lambda$ with $2d = 0.68 \text{ nm}$:

- First order: $\sin \theta = 0.154/0.68 = 0.226$, so $\theta = 13.1^\circ$.
- Second order: $\sin \theta = 0.453$, so $\theta = 26.9^\circ$.

(b) **Why X-rays.** Bragg reflection needs $\lambda \leq 2d = 0.68 \text{ nm}$. Visible light, at 400–700 nm, is about 1000 times too long: $\sin \theta$ would exceed 1, so there are no diffraction maxima at all, and the spacing is completely invisible. X-ray wavelengths (about 0.1 nm) are comparable to atomic spacings, so they give diffraction angles that are large and easy to measure. Franklin’s “Photo 51” showed a characteristic X-shaped pattern from DNA fibres, with layer lines related to the 0.34 nm stacking. That revealed a helix with a repeat of 3.4 nm, ten base pairs per turn.

P23.18

(a) **Perpendicular rays.** At Brewster’s angle, Snell’s law gives $n_1 \sin \theta_B = n_2 \sin \theta_2$, and Brewster’s condition gives $n_1 \sin \theta_B = n_2 \cos \theta_B$. Comparing the two:

$$\sin \theta_2 = \cos \theta_B \Rightarrow \theta_2 = 90^\circ - \theta_B.$$

The reflected ray leaves at θ_B on one side of the normal, and the refracted ray at θ_2 on the other side. The angle between them is $180^\circ - \theta_B - \theta_2 = 90^\circ$. ✓

Physically, the electrons in the glass oscillate along the direction of the refracted field. An oscillating dipole radiates nothing along its own axis, so light polarised in the plane of incidence cannot be reflected in the direction perpendicular to the refracted ray.

(b) Air into glass:

$$\theta_B = \tan^{-1} 1.50 = 56.3^\circ, \quad \theta_2 = 90^\circ - 56.3^\circ = 33.7^\circ.$$

Check: $\sin 56.3^\circ / \sin 33.7^\circ = 0.832 / 0.555 = 1.50$. ✓

(c) **Brewster windows.** Light polarised *in the plane of incidence* passes through a window at Brewster's angle with **no reflection loss** at all. Light polarised perpendicular to that plane loses roughly 15% per surface by reflection, on every pass. A laser beam bounces back and forth hundreds of times through the gain medium, so only the loss-free polarisation can build up and lase. The output is therefore almost completely linearly polarised.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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